t3-w02 — condition C1

Plane Shapes (2-D Shapes) — Plane shapes: types, properties and perimeter

> The objective plus the lesson plan's design: position, method labels, materials, entry behaviour and assessment.

INPUT LAYERS

Input layer	What it carries	Channel	In this condition
Scope	what is to be taught, and to whom	prompt	**present**
Prior knowledge	what the learner already knew coming in	prompt	**present**
Materials	what the lesson was represented with	prompt	**present**
Position	where this objective sat in the sequence	prompt	**present**
Method	the shape of the teaching, as event labels	prompt	**present**
Enacted content	what was actually explained and worked through	prompt	absent
Set work	what the learners were asked to do in class	prompt	absent
Assessment	the questions the lesson was checked with	prompt	**present**
Figures	the drawings the lesson used	tool	absent

WHAT THE MODEL IS SENT

Contract `ComposeWriteObjectiveSection`, declaring: `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Plane Shapes (2-D Shapes)

`subtopic`

Plane shapes: types, properties and perimeter

`term`

Third Term

`week`

2

`lesson_objectives`

• identify the types of plane shapes and state their properties, similarities and differences
• identify the parts of a circle
• find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

`entry_behaviour`

• Students have been taught angles.

`instructional_materials`

• Cardboard containing angles presentation.

Objective 1 — identify the types of plane shapes and state their properties, similarities and differences

`objective_text`

identify the types of plane shapes and state their properties, similarities and differences

`current_step`

1

`instructional_events`

• present_content
• elicit_performance
• present_content

`assessment_points`

• How many triangles has an octagon.

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 2 — identify the parts of a circle

`objective_text`

identify the parts of a circle

`current_step`

2

`instructional_events`

• present_content

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 3 — find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

`objective_text`

find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

`current_step`

3

`assessment_points`

- Find the perimeter of a square of sides 8cm
- Find the perimeter of a rectangle of length 7cm and breadth 4cm.
- What is the perimeter of regular pentagon with sides 5cm.
- Find the perimeter of a parallelogram whose base is 6cm and sides 5cm
- The circumference of a circle is 22cm – Find the radius of a circle $(\pi = \frac{22}{7})$.
- A rectangle is such that its length is twice its breadth, if the breadth is d, find the total length of the sides of rectangle.
- Calculate the perimeter of the figure below. [the figure is not included here, marked 1cm, 5cm, 9cm]
- Find the perimeter of a regular hexagon with sides 5cm.

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `instructional_events`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

WHAT THE MODEL PRODUCED

Objective 1

Types of Plane Shapes and Their Properties

EXPLANATION

A **plane shape** (or 2-D shape) is a flat figure that lies in a single plane.

- **Triangle** – 3 sides, 3 vertices, 3 interior angles. All triangles have the sum of interior angles equal to 180°.
- **Quadrilateral** – 4 sides and 4 vertices. Common types:
  - **Square** – 4 equal sides, 4 right angles, all sides parallel in opposite pairs.
  - **Rectangle** – opposite sides equal, 4 right angles.
  - **Parallelogram** – opposite sides equal and parallel, opposite angles equal.
  - **Trapezium (trapezoid)** – at least one pair of parallel sides.
- **Pentagon** – 5 sides, 5 vertices. Regular pentagon has equal sides and equal interior angles (108° each).
- **Hexagon** – 6 sides, 6 vertices. Regular hexagon: all sides equal, interior angles 120°.
- **Octagon** – 8 sides, 8 vertices. Regular octagon: all sides equal, interior angles 135°.
- **Circle** – not a polygon; it has no sides or vertices. It is defined by a centre point and a radius.

Similarities: All polygons (triangles, quadrilaterals, pentagons, etc.) are closed figures made of straight line segments. They all have as many interior angles as they have sides, and the sum of interior angles follows the rule \((n-2)\times180° \) where \(n\) is the number of sides.

Differences: The number of sides and vertices, the size of each interior angle, and whether sides are equal or parallel distinguish each shape.

WORKED_EXAMPLE

Worked Example – Comparing a Square and a Rectangle

Both shapes are quadrilaterals, but they differ in side lengths.

- A **square** has four equal sides (e.g., 5 cm each) and four right angles.
- A **rectangle** has opposite sides equal (e.g., 5 cm and 8 cm) and also four right angles.

Because all sides of a square are equal, a square is a special type of rectangle. However, not every rectangle is a square because the adjacent sides can be of different lengths.

1. 1. List three properties that are common to all quadrilaterals.
answer: They have four sides, four vertices, and the sum of interior angles is 360°.

2. 2. How many sides does a regular hexagon have, and what is each interior angle?
answer: A regular hexagon has 6 sides, and each interior angle is 120°.

3. 3. True or False: All rectangles are squares.
answer: False

4. 4. How many triangles can be formed by drawing all possible diagonals from one vertex of a regular octagon?
answer: 6 triangles (since an octagon can be divided into n-2 = 8-2 = 6 triangles).

Parts of a Circle

> **EXPLANATION**
> A circle is made up of several important parts:
>
> - **Centre**: The fixed point in the middle of the circle. Every point on the circle is the same distance from the centre.
> - **Radius**: A line segment that joins the centre to any point on the circle. All radii have the same length.
> - **Diameter**: A line that passes through the centre and whose endpoints lie on the circle. It is twice as long as the radius (Diameter = 2 × Radius).
> - **Circumference**: The distance around the circle, similar to the perimeter of a polygon.
> - **Arc**: A part of the circumference between two points on the circle.
> - **Chord**: A straight line joining any two points on the circle. If the chord passes through the centre, it is a diameter.
> - **Sector**: The region bounded by two radii and the arc between them (like a slice of pizza).
> - **Segment**: The region bounded by a chord and the arc it cuts off.
>
> Remember: every radius is equal, the centre is the same distance from any point on the circle, and the diameter is the longest chord.

1. Identify the part of the circle described: "A straight line that joins two points on the circle but does not pass through the centre."
answer: Chord

2. What is the relationship between the diameter and the radius of a circle?
answer: Diameter = 2 × Radius

3. If the radius of a circle is 5 cm, what is the length of the diameter?
answer: 10 cm

4. Name the part of the circle that is the distance around it.
answer: Circumference

5. A sector of a circle is bounded by two radii and an arc. If the radii are each 4 cm, what is the length of the two radii together?
answer: 8 cm

Objective 3

Finding Perimeters of Common Shapes

> EXPLANATION
> The perimeter of a shape is the total length around its edge. For the shapes you are revising, use these formulas:
>
> - **Square**: 4 \(\times\) side
> - **Rectangle**: 2 \(\times\) (length + breadth)
> - **Regular polygon** (all sides equal): number of sides \(\times\) side length
> - **Parallelogram**: 2 \(\times\) (base + side)
> - **Circle**: circumference = 2 \(\pi\) r, so if you know the circumference you can find the radius (r = circumference / (2\(\pi\))).
>
> Remember that "regular" means all sides and all interior angles are equal, so you only need the length of one side to find the perimeter.

> WORKED_EXAMPLE
> **Worked Example 1**
> Find the perimeter of a square whose side is 8 cm.
> Solution: Perimeter = 4 \(\times\) 8 cm = 32 cm.
>
> **Worked Example 2**
> Find the perimeter of a rectangle with length 7 cm and breadth 4 cm.
> Solution: Perimeter = 2 \(\times\) (7 cm + 4 cm) = 2 \(\times\) 11 cm = 22 cm.

1. Find the perimeter of a square with sides 8 cm.
answer: 32 cm

2. Find the perimeter of a rectangle with length 7 cm and breadth 4 cm.
answer: 22 cm

3. What is the perimeter of a regular pentagon with each side 5 cm?
answer: 25 cm

4. Find the perimeter of a parallelogram whose base is 6 cm and each side is 5 cm.
answer: 22 cm

5. The circumference of a circle is 22 cm. Find the radius of the circle (\(\pi = \frac{22}{7}\)).
answer: 1.5 cm

6. A rectangle has length twice its breadth. If the breadth is d cm, write an expression for the total length of all four sides of the rectangle.
answer: 6d cm

7. Find the perimeter of a regular hexagon with each side 5 cm.
answer: 30 cm

LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

Output layer	Made	Written from	Verdict
Title the reader-facing framing of the section	3	Scope	**grounded**
Explanation the principle, rule or relationship to review	3	Method written from event labels — the shape of the teaching without its substance	**substituted**
Worked example a problem with its solution worked through	2	nothing in the input	**unsourced**
Figure a drawing the learner reads from	0	nothing in the input	not produced

Practice what the learner does	16	Assessment written from the teacher's evaluation questions instead of her set work	**substituted**
Answer what the learner checks against	16	Assessment checked against the evaluation answers rather than the taught method	**substituted**